

Ontario Provincial Police

Ammunition

Purchase Requirement

PR-8310-OPP-3D-A

1. ITEM: 12 gauge Shotgun Ammunition, 70mm (2 ¾") length, Plastic Body, 438 grain (1oz.) Rifled Slug (Reduced Recoil) (Duty)

2. GENERAL REQUIRMENTS:

- a. Shall meet all requirements detailed in PR-8310-OPP-ALL

3. SPECIFIC REQUIREMENTS:

- a. Manufacturer shall complete Attachment A – Manufacture Data Sheet and return to O.P.P. with any submissions
- b. Crimp of shotgun shell shall be sufficient so as not to dislodge the projectile from its seated position in a previously un-chambered cartridge during recoil in O.P.P. firearms or their magazines
- c. The fired projectile shall not penetrate currently issued O.P.P. Level II or IIIA body armour listed below:
 - i. Level II – Pacific Safety Products,
 - 1. 33-010-0069 O.P.P. BALM (male)
 - 2. 33-010-0069 O.P.P. BALM (female)
 - ii. Level IIIA – Pacific Safety Products,
 - 1. 34-112-0071 (front)
 - 2. 34-113-0071 (rear)
- d. Projectile must be rifled slug construction. Desired effect that projectile consistently penetrate and average of 12" to 18" throughout the six (6) FBI gelatin tests, with no shots below 12", expand to maximum diameter and retain as high a percentage of its weight as possible. When fired out of a firearm as described in para 7b of this document
- e. Cartridge cap shall be manufactured from brass.
- f. It is preferred that the primer be sealed to prevent the entrance of moisture, which could alter the performance of components in various environmental conditions.

- g. Primer shall not be lead-free and be non-corrosive.

4. Packaging and identification

- a. Ammunition shall be contained in cartons of 5 cartridges per carton and cases of 250 or 500 cartridges.

5. Quality Control Plan/Quality Assurance Program

- a. See PR-8310-OPP-ALL for details and Deliverables 1.5.6 Vendor Quality Control Plan and Quality Control Program.

6. Essential Quality Standards

- a. VELOCITY – An average of 10 at 3 ft.: 1250 f/s (+/- 90 f/s) when tested from a SAAMI 30" test barrel. Based on ANSI/SAAMI Z299.2-2019
- b. PRESSURE – A maximum average of 10: 11,500 P.S.I. using piezoelectric system for pressure measuring and SAAMI test equipment. Based on ANSI/SAAMI Z299.2-2019
- c. PRIMER – Standard lead styphnate primer must be used, it is preferred that the primer be sealed
- d. DEFECTIVE TOLERANCES – Ammunition not falling within the noted tolerances may be subject to return to the supplier together with all the ammunition from the same lot for replacement or monetary return at the discretion of the O.P.P. Ammunition returned under this condition will be at the supplier's cost.

ANSI Standard on Inspection by Attributes ANSI/ASQ Z-1.4-2003 (R2013), Table I-Sample size code letters, General inspection level II, Table III-A Double Sampling Plan for normal inspection (Master table).

DEFECT DESCRIPTION	ALLOWABLE QUALITY LEVEL
Critical (Functional)	0.00

Any ammunition manufacturing defect which would result in cartridge failure producing a firearms blow-up. Any loading or propellant powder defect which may result in a projectile, wad or piece of shot column lodging in the bore of a firearm.

DEFECT DESCRIPTION	ALLOWABLE QUALITY LEVEL
Major (Functional)	0.010

Misfires, hard extractions or any other defect which would seriously alter functioning or performance.

DEFECT DESCRIPTION	ALLOWABLE QUALITY LEVEL
Major (Visual & Dimensional)	0.010

Any visual or dimensional defect which would seriously alter functioning or performance.

DEFECT DESCRIPTION	ALLOWABLE QUALITY LEVEL
Minor (Visual & Dimensional)	0.15

Any visual or dimensional defect which would not seriously alter functioning or performance.

7. Performance Standards

a. Accuracy Specification

Five – 5 shot groups must average not more than 10.6 cm (3”) at 22.86 m (25 yds) fired from a Remington 870 Police Magnum Shotgun with a 50.8 cm (20”) cylinder bore barrel, hand held on a sand bag rest.

and/or

A 25 shot group fired from a universal receiver mounted test barrel manufactured to SAAMI standards for a 12g, 2 ¾” shall be less than 7.62 cm (3.5”) at 22.86 m (25 yds).

Ammunition not meeting this accuracy specification will be treated as nonconforming the stated tolerance.

- b. Ammunition must be compatible (feed and function) with the following firearms
 - i. Remington 870 Police Magnum Shotgun, 12g 70 mm (2 $\frac{3}{4}$ ") and 75 mm (3"), with 50.8 cm (20") cylinder bore barrel
 - ii. Remington 870 Wingmaster Shotgun, 12g 70 mm (2 $\frac{3}{4}$ ") and 75 mm (3"), with 50.8 cm (20") cylinder bore barrel

8. Identifications Standards

- a. See PR-8310-OPP-ALL for details with addition(s) detailed in Para 4 of individual specifications.

9. Quality Assurance

- a. See PR-8310-OPP-ALL for details

ATTACHMENT A

Manufacture Data Sheet

**12 gauge Shotgun Ammunition, 70mm (2 ¾") length, Plastic Body, 438 grain
(1oz.) Rifled Slug (Reduced Recoil) (Duty)**

Manufacturer:			
Principal Technical Contact:			
Telephone:			
Email address:			
Caliber/Gauge:	12g, 70 mm (2 ¾") rifled slug		
Product name:		Purpose:	Duty
Product Order/Part #:		Sub-Purpose:	General Duty
Rounds per box:			
Boxes per case:			
Projectile composition:			
Ballistic Coefficient:			
Sectional Density:			
LOT/DAY CODE numbers:	Lot 1:		Lot 2:
Meaning of characters in LOT/DAY CODE numbers:			

Component compositions			
ITEM	MANUFACTURER	TYPE	WEIGHT (if applicable)
Projectile:			
Case mouth Sealant:			
Case:			
Primer:			
Primer Sealant:			
Propellant:			
Headstamp marking:			
Headstamp meaning:			
Material Safety Data sheets attached:			
Production group size standard measured at 22.86 m (25 yds):		Inches	# of rounds fired:
Recommended velocity window for optimal terminal performance in ballistic gelatin (fps):	Minimum:		Maximum:

<u>Exterior ballistics data based on manufacturer's averages–</u>		<u>Remington 870</u>	<u>w/ 22.86 m (25 yd) zero</u>	
Distance	Velocity (fps)	Energy (foot pounds)	Trajectory (inches)	
Muzzle				
4.57 m (5 yds)				
9.14 m (10 yds)				
13.72 m (15 yds)				
18.29 m (20 yds)				
22.86 m (25 yds)				
45.72 m (50 yds)				
Maximum Range		Yards @		Angle

<u>Instrumental 91.44 cm (3') Velocity Parameters – Target Velocity/Acceptable Velocity Data</u>			
Target velocity average:	Lowest acceptable target velocity	Highest acceptable target velocity	Highest acceptable velocity extreme spread

Following only required for “Duty” submissions.

<u>Penetration - Manufacturers Gelatin Testing Data</u>				
Description	Penetration	Standard Deviation	Expansion	Retained weight (%)
Average from all 6 tests (30 rounds - Not averages of averages)				
INDIVIDUAL TESTS (5 rounds per event)				
Bare gelatin				
Heavy clothing				
Steel				
Wallboard				
Plywood				
Automobile glass				
Total shots <30.48 cm (12")				